

# CLMS Product Filenaming - Design Principles

Copernicus Land Monitoring Service - Technical  
Library



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# 1. CLMS Product Filenaming - Design Principles

Grounded in the parsEO schema registry, CDSE product catalogue, and current filenaming conventions.

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## 1.1 Purpose and Scope

These principles govern the filenaming of **all new CLMS product deliveries** - new products, reprocessings, and new versions of existing products. They describe the target convention; existing files on CDSE under legacy names remain as-is but are non-compliant.

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## 1.2 Principle 1 - Filename Structure: fields, delimiter, extension

**Rule:** A CLMS filename consists of a stem (base name) and if required (e.g. not folder names) an extension, separated by a period .:

```
{stem}.{extension}
```

The stem is composed of **fields** - descriptive elements separated by underscore \_. Each field carries a specific meaning:

```
field1_field2_field3_..._fieldN.ext
```

Fields may not be empty. A field delimiter may not appear at the beginning or end of a filename, nor twice in a row.

**Extension:** The file extension (.tif, .gpkg, .xml) indicates the file format and is not part of the naming convention.

**Compliant:** CLMS\_VLCC-GRA\_S2021\_R10m\_E27N48\_3035\_V01-R00.tif

**Non-compliant:** \_CLMS\_VLCC-GRA\_... (leading delimiter), CLMS\_VLCC-GRA\_\_S2021\_... (double delimiter)

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## 1.3 Principle 2 - Allowed Characters: uppercase, digits, underscore, hyphen

**Rule:** Filenames MUST use only the following ASCII characters:

| Character | ASCII | Usage                                 |
|-----------|-------|---------------------------------------|
| A-Z       | 65-90 | Uppercase letters only (no lowercase) |
| 0-9       | 48-57 | Digits                                |
| _         | 95    | Field delimiter                       |
| -         | 45    | Within-field separator                |

The period . (ASCII 46) is reserved exclusively for the file extension separator. Spaces, lowercase letters, and all other special characters are NOT permitted.

**Rationale:** A restricted character set enables deterministic, case-insensitive parsing across all platforms. Uppercase-only avoids case-sensitivity issues between Linux (case-sensitive) and Windows/macOS (case-insensitive).

**Compliant:** CLMS, V01-R00, 054-0154-IW1, R20-5m, VLCC-GRA

**Non-compliant:** clms\_EGMS, V01\_r00, P20x5m, data file.tif

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## 1.4 Principle 3 - Maximum Filename Length

**Rule:** The complete filename (stem + extension) MUST NOT exceed 255 characters. The fully qualified file name (including path) SHOULD NOT exceed 260 characters.

**Recommendation:** Keep filenames under 100 characters to safely accommodate both file name and storage path across all platforms.

**Rationale:** The 255-character limit is the smallest common denominator across Windows (260 char FQFN), macOS (1024), and Linux (4096). The 100-character recommendation preserves room for deep directory structures.

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## 1.5 Principle 4 - Delimiter: Underscore \_ between fields, hyphen - within fields

**Rule:** Use \_ to separate all top-level fields. The hyphen - is the **within-field separator**: it joins the variable (and sub-variables) to the product code (VLCC-GRA, WSI-SP-SCD), joins version to revision (V01-R00), and appears where a notation itself requires it (e.g., 054-0154-IW1 for IW burst tile identifiers, C2018-2021 for change periods). It is never a cross-field delimiter.

**Rationale:** Consistent underscore delimiters make tokenisation deterministic across all product families. The hyphen carries all intra-field structure, so splitting on \_ always yields the same field count within a product family.

**Compliant:**

- CLMS\_VLCC-GRA\_S2021\_R10m\_E27N48\_3035\_V01-R00.tif
- CLMS\_EGMS-L2a-A\_T20190101P5Y\_R20-5m\_054-0154-IW1\_V01-R00.tif (hyphens inside EGMS-L2a-A and 054-0154-IW1 are semantic)
- CLMS\_CLCPLUS-LCU\_S2023\_R10m\_E48N37\_3035\_V01-R00.tif

**Edge case:** Hyphenated geographic names (e.g., "Bourg-en-Bresse") is a single token; the hyphen is part of the name, not a delimiter.

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## 1.6 Principle 5 - Field Order: Variable hyphen-appended to the product code, token count invariant

**Rule:** Fields appear in a fixed order per product family. The variable is **not a standalone field**: it is hyphen-appended to the product code in position 2 ({CODE}-{VARIABLE}[-{SUBVARIABLE}]). Underscore-separated token count is invariant across a product family - a sub-product identifier is never a separate positional field; it is encoded within the code-variable compound:

**Canonical template:**

CLMS\_{CODE}-{VARIABLE}[-{SUBVARIABLE}][\_ {TEMPORAL}]{RES}\_{EXTENT}\_{EPSG}\_{VERSION}-{REVISION}

Sub-products within a family (e.g., SP, SWS, WDS within WSI) extend the compound with a further hyphen (WSI-SP-SCD, WSI-WDS-SSC). This keeps the underscore token count identical for every file in the family.

**Rationale:** Code and variable together identify **what** the file is; the remaining fields identify **when, where, and which version**. Placing the full product identity at position 2 groups all files of a product and its layers together in directory listings, and a parser always knows which position holds which field - no branching logic based on which sub-product happens to be present.

**Compliant:**

- CLMS\_VPP2-ST-PPI\_A20240101P10D\_R10m\_T33UVS\_V01-R00.tif
- CLMS\_WSI-SP-SCD\_A20200901P1Y\_R20m\_T38TKL\_V01-R00.tif
- CLMS\_WSI-WDS-SSC\_20210217T053159\_R60m\_T32TNS\_S1B\_V01-R00.tif (sensing datetime, sub-product in the code-variable compound)

## 1.7 Principle 6 - Programme Prefix: CLMS\_ mandatory, no exceptions

**Rule:** Every CLMS product filename MUST begin with CLMS\_. No exceptions. No product is exempt. This applies to all new deliveries, reprocessings, and new product versions across the entire CLMS portfolio (EEA, JRC, contractors).

**Rationale:** The prefix makes programme identity parseable from the filename alone. Any file without CLMS\_ is immediately identifiable as non-conformant.

**Compliant:**

- CLMS\_VLCC-GRA\_S2021\_R10m\_E27N48\_3035\_V01-R00.tif
- CLMS\_VPP2-ST-PPI\_A20240101P10D\_R10m\_T33UVS\_V01-R00.tif
- CLMS\_UA-LCU\_S2021\_V025ha\_DK004L3\_3035\_V01-R00.gpkg
- CLMS\_CLCPLUS-LCU\_S2018\_R10m\_EUROPE\_3035\_V01-R00.tif
- CLMS\_CLC-LCU\_S2018\_V25ha\_EUROPE\_3035\_V01-R00.gpkg
- CLMS\_EGMS-L3-U\_T20190101P5Y\_R100m\_E28N49\_3035\_V01-R00.tif

## 1.8 Principle 7 - Product Code: Short, uppercase, registered

**Rule:** Product codes must be registered in the CLMS product code registry before schema authoring. Uppercase alphanumeric only. Prefer 4–7 characters. Retain the HR prefix only when it is part of the established acronym.

**Compliant:** VLCC, VPP2, WSI, EUHYDRO, CLCPLUS, UA

**Non-compliant:** HRVPP2 (redundant HR), VPP-2 (hyphen in code), any unregistered code

## 1.9 Principle 8 - Temporal Coverage: Format matches product type

**Rule:**

| Situation       | Format              | Examples        | Definition                                                  |
|-----------------|---------------------|-----------------|-------------------------------------------------------------|
| Sensing instant | {YYYYMMDD}T{HHMMSS} | 20210217T053159 | Single acquisition at a specific date+time (e.g. satellite) |

| Situation  | Format                 | Examples                                        | Definition                                                                                                                                                                                                                                                                                         |
|------------|------------------------|-------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|            |                        |                                                 | scene). Zero-duration temporal instant. Uses bare ISO 8601 <code>dateTime</code> lexical form - the <code>T</code> separator and fixed-width format make it self-identifying, so no semantic prefix is needed.                                                                                     |
| Aggregated | A{YYYYMMDD}P{duration} | A20240101P10D,<br>A20240101P1M,<br>A20240101P1Y | A 2D snapshot where pixel values are derived by <b>compositing, averaging, or selecting</b> from multiple observations over time using a defined algorithm. The temporal dimension is <b>collapsed</b> to a single representative value per pixel; the file is not designed for temporal querying. |
| Status     | S{YYYY}                | S2021                                           | A static thematic map or interpreted classification representing a single reference year. The product does <b>not</b> represent a statistical aggregate of repeated measurements; it is the result of a single survey, interpretation, or classification exercise.                                 |
| Change     | C{YYYY}-{YYYY}         | C2018-2021                                      | A map of transitions or differences between two Status reference years. The two years are the start and end of the change period. The output is a 2D change layer (not a time-series stack).                                                                                                       |
| Timeseries | T{YYYYMMDD}P{duration} | T20190101P5Y                                    | A single file containing <b>preserved multi-temporal measurements</b> where the temporal dimension is queryable (e.g. as internal bands, layers, or a time-enabled data cube). Designed for temporal analysis, not as a collapsed 2D snapshot.                                                     |
| Forecast   | F{YYYYMMDD}P{duration} | (Reserved)                                      | <b>Reserved/Proposed.</b> For future forecast/modelled products. Not currently in use.                                                                                                                                                                                                             |

**Temporal reference standard:** All temporal formats follow **ISO 8601-1:2019** (Date and time - Representations for information interchange). The `P{duration}` suffix uses the ISO 8601 duration format (`P10D`, `P1M`, `P1Y`). Reference: <https://www.iso.org/standard/70907.html>

**Edge case:** Multi-seasonal products encode the season in the code-variable compound (`-S1`, `-S2`), not in the temporal field. This is consistent with external standards (CF Conventions, NASA EOSDIS) which treat season as a data attribute, not a temporal primitive.

**Compliant examples:**

- `CLMS_VPP2-ST-PPI_A20240101P10D_...` (aggregated, dekadal)
- `CLMS_VLCC-GRA_S2021_...` (annual status)
- `CLMS_UA-LCU_C2018-2021_...` (change between two Status years)
- `CLMS_WSI-WDS-SSC_20210217T053159_R60m_T32TNS_S1B_V01-R00.tif` (sensing datetime)
- `CLMS_EGMS-L3-U_T20190101P5Y_R100m_E28N49_3035_V01-R00.tif` (timeseries, multi-year)

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## 1.10 Principle 9 - Resolution: `R{value}[unit]` for raster, `V{value}[unit]` for vector and point grid

**Rule:** Raster and gridded products use `R{value}[unit]` with no zero-padding: `R10m`, `R1km`, `R100m`. The value is the integer pixel/grid spacing in metres. Irregular grid spacings join both values with a hyphen within the field: `R20-5m` (e.g., EGMS L2 20×5 m posting).

The `V` prefix covers two cases:

- **Vector minimum mapping units** - `V{value}ha` (e.g., `V025ha` for Urban Atlas 0.25 ha, `V25ha` for CLC 25 ha). Sub-1 ha values use a leading zero with no decimal point.
- **Point grid posting spacing** - `V{value}m` for regular grids (e.g., `V100m` for EGMS L3) or `V{value}-{value}m` for irregular grids (e.g., `V20-5m` for EGMS L2 point data)

Only `R` and `V` exist - there is no separate point-grid prefix.

**Compliant:** `R10m`, `R1km`, `R100m`, `R20-5m`, `V025ha`, `V25ha`, `V100m`, `V20-5m`

**Edge case:** Products with no meaningful spatial resolution (e.g., nation-level statistical vector layer) omit the resolution token.

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## 1.11 Principle 10 - Spatial Extent: Identifies the area covered by this specific file

**Rule:** This field identifies the spatial extent of the individual file - not the overall product coverage. It can be a grid tile, a satellite scene, an administrative boundary, or a continental code:

| Extent type   | Description                                         | Examples     |
|---------------|-----------------------------------------------------|--------------|
| MGRS          | Sentinel-2 raster tile                              | T33UVS       |
| LAEA 100km    | Pan-European composite on INSPIRE/ ETRS89-LAEA grid | E27N48       |
| IW burst      | Sentinel-1 swath (burst-level)                      | 054-0154-IW1 |
| FUA code      | Functional Urban Area, city-level vector            | DK004L3      |
| Country code  | Nation-level vector                                 | DE, IT       |
| Delivery Unit | Tiling system for large-area products               | DU001        |

| Extent type   | Description                              | Examples                                  |
|---------------|------------------------------------------|-------------------------------------------|
| Continental   | Full European continent coverage         | EUROPE                                    |
| Hemispherical | Global product subset for one hemisphere | NORTHERNHEMISPHERE,<br>SOUTHERNHEMISPHERE |
| Global        | Full global coverage                     | GLOBAL                                    |

Spatial extent identifiers are \_-separated from adjacent tokens. Never fuse them with EPSG or resolution.

**Note:** This field describes the extent of the **file**, not the product. A pan-European product may be delivered as individual MGRS tiles, each with a different spatial extent. Likewise, **EUROPE** means this specific file covers the entire continent - it does not mean the product is “pan-European” by design.

**Compliant:**

- CLMS\_VPP2-ST-PPI\_...\_T33UVS\_... (MGRS tile)
- CLMS\_VLCC-GRA\_S2021\_R10m\_E27N48\_... (LAEA tile)
- CLMS\_CLCPLUS-LCU\_S2023\_R10m\_E48N37\_... (LAEA tile)
- CLMS\_UA-LCU\_S2021\_V025ha\_DK004L3\_... (FUA code)
- CLMS\_EGMS-L2a-A\_...\_054-0154-IW1\_... (IW burst)
- CLMS\_EUHYDRO-NET-LINE-EUROPE\_... (continental coverage)
- CLMS\_WSI-WDS-SSC\_20210217T053159\_R60m\_T32TNS\_S1B\_V01-R00.tif (MGRS, SAR sensor)

## 1.12 Principle 11 - EPSG Code: Bare code, required for projected CRS, omit when CRS is implicit

**Rule:** Standalone \_-separated token containing the bare EPSG code with **no zero-padding** (3035). Required for products with a projected CRS (e.g., LAEA 100km extents). Omit when the CRS is implicit in the spatial extent system (e.g., MGRS tiles imply UTM, so EPSG is redundant). Also omit for CRS-independent files (metadata XML, archive containers).

**Rationale:** The bare code is the EPSG identifier exactly as registered (<https://epsg.org>) - 3035, 4326, 32633 - so the token matches the authority and needs no transformation for lookups. EPSG codes are 4–5 digits; the field is bounded, and the surrounding fixed field order keeps parsing deterministic without artificial padding.

**Compliant:**

- CLMS\_VLCC-GRA\_S2021\_R10m\_E27N48\_3035\_V01-R00.tif (LAEA tile, EPSG 3035 required)
- CLMS\_CLCPLUS-LCU\_S2023\_R10m\_E48N37\_3035\_V01-R00.tif
- CLMS\_VPP2-ST-PPI\_A20240101P10D\_R10m\_T33UVS\_V01-R00.tif (MGRS tile - EPSG omitted, CRS implicit)

**Non-compliant:** 03035 (zero-padded), EPSG3035 (prefixed)

## 1.13 Principle 12 - Version: V{XX}-R{YY} single hyphenated field

**Rule:** A single compound token: V{XX} (major, 2-digit zero-padded), hyphen, R{YY} (revision, 2-digit zero-padded). Never use packed V{XXX}.

**Compliant:** V01-R00, V01-R01, V02-R00

## 1.14 Principle 13 - Variable: Hyphenated composite appended to the product code

**Rule:** The variable identifies the measured or mapped quantity. It is hyphen-appended to the product code at position 2, forming the product identity compound:

```
{CODE}-{PRODUCT_TYPE}[-{PARAMETER}][-{SEASON}][-QFLAG]
```

- **PRODUCT\_TYPE:** layer type code (ST, VPP, GPP, SP, LCU)
- **PARAMETER:** specific measurement (PPI, SOSD, TOTAL, SCD)
- **SEASON:** S1 or S2 for multi-seasonal products
- **QFLAG:** literal suffix for quality flag layers - always last

LCU (Land Cover / Land Use) is the shared variable for all status and change land cover/use products (UA, CLC, CLCPLUS, RZ, ...); status vs change is expressed by the temporal field (S{YYYY} vs C{YYYY}-{YYYY}), not by the variable.

**Compliant:**

|                                |                          |
|--------------------------------|--------------------------|
| VPP2-ST-PPI                    | → VPP2-ST-PPI-QFLAG      |
| VPP2-VPP-SOSD-S1               | → VPP2-VPP-SOSD-S1-QFLAG |
| VPP2-GPP-TOTAL                 | → VPP2-GPP-TOTAL-QFLAG   |
| WSI-SP-SCD                     | → WSI-SP-SCD-QFLAG       |
| UA-LCU / CLC-LCU / CLCPLUS-LCU |                          |

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## 1.15 Principle 14 - Production Date: Metadata-only unless operationally required

**Rule:** Omit production date from filenames. Include it only when multiple production runs for the same temporal period coexist in the same distribution channel. When present: {YYYYMMDD} as the last token before the extension.

**Compliant:** CLMS\_EUHYDRO-NET-LINE\_EUROPE\_3035\_V02-R00\_20261015.gpkg

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## 1.16 Principle 15 - Sensor Token: Omit unless multi-sensor discrimination is required

**Rule:** No standalone sensor token in the filename. The sensor is implicit in the product code. Include a sensor token only when the product supports multiple sensor sources distinguishable at file level (e.g., SAR platform identity \_S1B\_).

**Compliant:**

- CLMS\_WSI-SP-SCD\_A20200901P1Y\_R20m\_T38TKL\_V01-R00.tif
- CLMS\_WSI-WDS-SSC\_20210217T053159\_R60m\_T32TNS\_S1B\_V01-R00.tif (S1B sensor token required for SAR discrimination)

## 1.17 Principle 16 - Quality Layers: Separate file, hierarchical suffix -QFLAG

**Rule:** Quality layers are separate files. Same stem as the data layer, with -QFLAG appended to the appropriate hierarchical level of the code-variable compound. No other field may differ between a data file and its quality file.

The -QFLAG suffix attaches at the level it applies to:

- VPP2-VPP-SOSD-QFLAG - quality applies to the VPP-SOSD product only
- VPP2-VPP-QFLAG - quality applies to all VPP sub-products (SOSD, TOTAL, SEASONAL, etc.)
- VPP2-ST-PPI-QFLAG - quality applies to the PPI measurement only

**Compliant:**

```
CLMS_VPP2-ST-PPI_A20240101P10D_R10m_T33UVS_V01-R00.tif ← data
CLMS_VPP2-ST-PPI-QFLAG_A20240101P10D_R10m_T33UVS_V01-R00.tif ← quality
```

## 1.18 Principle 17 - Schema Registration: parseEO schema BEFORE CDSE onboarding

**Rule:** Every new CLMS product must have a registered product code, a parseEO schema with `status: current`, and a naming crosswalk before the first CDSE delivery. The schema defines the filename; CDSE filenames are not reverse-engineered into schemas after the fact.

## 1.19 What Cannot Be Standardised

Some aspects genuinely differ between product families and should not be forced into a single mould:

1. **Raster vs. vector/point-grid resolution** - R10m and V25ha are different physical quantities; V covers both vector MMU (V{value}ha) and point grids (V{value}m)
2. **Pan-European vs. local tile encoding** - LAEA grid tiles (E27N48) for pan-European products, FUA codes (DK004L3) for city-level vector - these are incompatible spatial encodings, both valid
3. **Annual status vs. scene-level temporal format** - S{YYYY} for static maps, {YYYYMMDD}T{HHMMSS} for single acquisitions - the right format depends on the product's temporal nature
4. **Multi-resolution families** - the set of valid resolution values varies per product; each schema enumerates its own
5. **Production date inclusion** - reflects operational policy; included when corrections under the same version are expected

## 1.20 Principles Summary

| # | Principle           | Rule                                                                                    |
|---|---------------------|-----------------------------------------------------------------------------------------|
| 1 | Filename structure  | Fields separated by _, stem.extension                                                   |
| 2 | Allowed characters  | A-Z, 0-9, _, - only; no lowercase, no spaces                                            |
| 3 | Max filename length | ≤255 chars stem+extension, recommend ≤100                                               |
| 4 | Delimiter           | _ between fields; - within fields (code-variable compound, version-revision, notations) |

| #  | Principle       | Rule                                                                                                                      |
|----|-----------------|---------------------------------------------------------------------------------------------------------------------------|
| 5  | Field order     | Variable hyphen-appended to product code at position 2; canonical template defines field sequence                         |
| 6  | Prefix          | CLMS_ mandatory for all products, no exceptions                                                                           |
| 7  | Product code    | Short, uppercase, registered in code registry                                                                             |
| 8  | Temporal        | A{start}P{duration} (aggregated), S{YYYY} (status), C{YYYY}-{YYYY} (change), T{start}P{duration} (timeseries)             |
| 9  | Resolution      | R{XX}m (raster/grid), V{value}ha (vector MMU), V{value}m (point grid) - hyphen for irregular grids                        |
| 10 | Spatial extent  | MGRS / LAEA / IW burst / FUA / country / DU / EUROPE - identifies the area of the specific file, not the product coverage |
| 11 | EPSG            | Bare EPSG code, no zero-padding (3035); required for projected CRS, omit when CRS is implicit in the spatial extent       |
| 12 | Version         | V{XX}-R{YY} single hyphenated field; no packed V{XXX}                                                                     |
| 13 | Variable        | {CODE}-{TYPE}[-{PARAM}][-{SEASON}][-QFLAG] compound at position 2                                                         |
| 14 | Production date | Metadata-only unless operationally required                                                                               |
| 15 | Sensor          | Omit unless multi-sensor discrimination needed                                                                            |
| 16 | Quality         | Separate file, hierarchical -QFLAG suffix, identical stem                                                                 |
| 17 | Registration    | parsEO schema before CDSE onboarding                                                                                      |

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*Document status: Draft for review. Grounded in the parsEO schema registry and VPP2 naming crosswalk.*